Blindness Pre

Ophthalmology | 3rd Year MBBS

Hi Sammy!
Blindness prevention =
PSM + Ophthal cross Q!
Statistics are KEY!

Blindness | NPCBVI | Vision 2020 | Ophthalmology

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WHO definitions are
asked in EVERY exam!
Know the VA cutoffs!

WHO Definitions (ICD-11, 2019 Revision)

Category	VA (Better Eye)	Additional Criteria
Normal vision	>= 6/12	No impairment
Mild VI	< 6/12 to >= 6/18	Not included in WHO VI definition
Moderate VI	< 6/18 to >= 6/60	Presenting VA, better eye
Severe VI	< 6/60 to >= 3/60	Presenting VA, better eye
Blindness	< 3/60	OR visual field < 10 degrees

★ **Indian NPCB definition: Presenting VA < 6/60 in better eye (earlier used < 6/60).**
OR visual field < 20 degrees from center of fixation in better eye.
Note: Indian definition is slightly different from WHO — leads to higher prevalence figures.

Global Burden of Blindness (WHO 2021)

Fact	Data
Total people with vision impairment	~2.2 billion worldwide
People who are blind globally	~43 million
Avoidable blindness	~80% of all visual impairment
Leading global cause	Uncorrected refractive errors (#1), Cataract (#2)
Most affected regions	South Asia, Sub-Saharan Africa

★ **Avoidable Blindness**
Avoidable = Preventable + Treatable. 80% of global blindness is avoidable! (WHO)

Cataract = #1 cause of blindness in India!
This data = exam gold!

Causes of Blindness — NBVI Survey 2015-19

#	Cause	%	Key Point
1	Cataract	66.2%	#1 cause, treatable
2	Corneal opacity	8.2%	#1 in age < 50 years
3	Cataract surgical complications	7.2%	Post-op complications
4	Posterior segment disorders	5.9%	DR, ARMD, RD
5	Glaucoma	5.5%	Irreversible damage
6	Refractive errors	~3%	#1 cause globally
7	Others	~4%	Optic atrophy, amblyopia

India — Key Blindness Statistics

Parameter	Value
Prevalence of blindness (2015-19)	0.36% (reduced from 1.1% in 2001-02)
Prevalence of visual impairment	2.55%
Avoidable blindness proportion	92.9% of all blindness
Cataract surgical rate (CSR)	~6000 per million population per year
Annual cataract surgeries	~6.5 million (highest globally!)
Reduction achieved	47.1% reduction from 2001-02 baseline

Age > 50 years: #1 cause = Cataract (66.2%)

Age < 50 years: #1 cause = Corneal opacity (37.5%), followed by refractive errors

Children: Congenital anomalies, Vitamin A deficiency, ROP (premature infants)

NPCBVI = India's oldest
eye health programme!
Year + objectives = HY!

National Programme for Control of Blindness + VI

Launched:	1976 (India = FIRST country to have national blindness programme)
Renamed:	NPCBVI in 2017 (Visual Impairment added)
Funding:	100% Centrally Sponsored Scheme initially; now 60:40 (Centre:State)
NE states:	90:10 funding ratio
Goal:	Reduce prevalence of blindness from 1.4% to 0.3%
NHP target:	Reduce blindness to 0.25% by 2025
CSR target:	6000 cataract surgeries per million population per year
Strategy:	Free cataract surgery + IOL, school eye screening, Vit A programme

Components of NPCBVI

Component	Detail
Cataract surgery	Free IOL surgery in govt hospitals + camps
School eye screening	Refractive error correction, free spectacles
Vitamin A programme	Supplementation for children 9 months - 5 years
Eye banking	Collection, storage, distribution of donor corneas
Glaucoma control	Early detection + treatment camps
Diabetic retinopathy	Screening of diabetic patients, laser facilities
Childhood blindness	ROP screening, congenital cataract surgery
Training / HRD	Training ophthalmologists, ophthalmic assistants

Vision 2020 + Survey Data

Vision 2020 = WHO + IAPB global initiative! Know the priorities listed!

Vision 2020: The Right to Sight

Launched:	1999 by WHO + IAPB (International Agency for Prevention of Blindness)
Goal:	Eliminate avoidable blindness by 2020
Pillars:	1. Disease control 2. HRD 3. Infrastructure + technology
Priority diseases:	Cataract, Trachoma, Onchocerciasis, Childhood blindness, RE

Vision 2020 Priority Conditions

Condition	Strategy
Cataract	Increase CSR, IOL implantation, quality surgery
Trachoma	SAFE strategy (Surgery, Antibiotics, Face washing, Environment)
Childhood blindness	Vit A, ROP screening, congenital cataract surgery
Refractive errors	School screening, affordable spectacles
Low vision	Rehabilitation services, assistive devices
Glaucoma	Early detection, IOP monitoring, access to treatment
Diabetic retinopathy	DR screening, laser therapy, anti-VEGF access

Blindness Surveys in India

Survey	Year	Key Finding
1st National Survey	1986-89	Prevalence: 1.49%
Rapid Survey	2001-02	Prevalence: 1.1%
Rapid Survey	2006-07	Prevalence: 1.0%
NBVI Survey	2015-19	Prevalence: 0.36% (major reduction!)

Blindness stats = easy marks if memorized! You've GOT this Sammy!

Most Repeated LAQs (10-12 marks)

- 1. Describe NPCBVI — history, objectives, components, achievements
- 2. Enumerate causes of blindness in India. Describe prevention strategies
- 3. Write about Vision 2020 — objectives, priority diseases, pillars
- 4. Define blindness (WHO). Describe national surveys and their findings

Most Repeated SAQs (5-6 marks)

- 1. WHO definition of blindness and visual impairment
- 2. Causes of blindness in India with percentages
- 3. SAFE strategy for trachoma control
- 4. Cataract surgical rate — definition and India's target
- 5. School eye screening programme
- 6. Eye banking in India

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
#1 cause of blindness in India	Cataract (66.2%)
#1 cause globally	Uncorrected refractive errors
WHO blindness definition	VA < 3/60 in better eye
NPCB launched year	1976 (renamed NPCBVI in 2017)
Vision 2020 launched by	WHO + IAPB in 1999
Current blindness prevalence India	0.36% (NBVI Survey 2015-19)
India CSR target	6000 per million per year
Avoidable blindness %	~80% globally, ~93% in India
SAFE strategy for	Trachoma elimination
First country with blindness programme	India (NPCB, 1976)
NHP blindness target	0.25% by 2025

♥ India leads the world in cataract surgeries, Sammy! Public health saves millions of eyes. You're going to be part of this incredible mission!